

HW#6

Finite Wings

1. Calculate the induced velocity for a parabolic circulation distribution:

$$\Gamma = \Gamma_0 \left(1 - \frac{y^2}{(b/2)^2} \right)$$

2. The measured lift slope for the NACA 23012 airfoil is $0.1080 \text{ degree}^{-1}$, and $\alpha_{L=0} = -1.3^\circ$. Consider a finite wing using this airfoil, with $AR=8$ and taper ratio $= 0.8$. Assume that $\delta=\tau$.

Calculate the lift and induced drag coefficients for this wing at a geometric angle of attack $= 7^\circ$.

3. The piper Cherokee (a light, single-engine general aviation aircraft) has a wing area of 170 ft^2 and a wing span of 32 ft. Its maximum gross weight is 2450 lb. The wing uses an NACA 65-415 airfoil, which has a lift slope of $0.1033 \text{ degree}^{-1}$ and $\alpha_{L=0} = -3^\circ$. Assume $\tau = 0.12$. If the airplane is cruising at 120 mi/h at standard sea level at its maximum gross weight and is in straight and level flight, calculate the geometric angle of attack and induced drag ($e=0.64$) of the wing.

4. Consider a planar wing (i.e., no geometric twist) which has a NACA 0012 section and an aspect ratio of 7.0. Use a four-term series to represent the load distribution. Compare the lift coefficient, the induced drag coefficient, and the spanwise lift distribution for taper ratios of (a) 0.4, (b) 0.5, (c) 0.6, and (d) 1.0. Hint: Read EXAMPLE 7.2 of the book.